Guidance document on Antimicrobial Susceptibility Testing of *Legionella pneumophila*

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Introduction

Legionella pneumophila is a facultative intracellular, aerobic, gram-negative bacillus, and the etiological agent of Legionnaires' disease. Due to its fastidious nature, antimicrobial susceptibility testing is difficult and the correlation between in vitro results and clinical outcome is uncertain.

Antimicrobial resistance

Acquired antimicrobial resistance in *L. pneumophila* is rare [1-6]. However, a clinical isolate resistant to ciprofloxacin due to a single point mutation in the *gyrA* gene has been reported [7] and fluoroquinolone resistance have been described from patients during therapy [8]. An efflux pump (LpeAB) has been demonstrated to be responsible for elevated MICs (0.125-2 mg/L) for azithromycin [1]. The efflux pump is found in clinical isolates of *L. pneumophila*, related to specific sequence types [1,4,9]. Any clinical significance of reduced susceptibility to azithromycin has not been demonstrated. These findings also suggest erythromycin to be a poor predictor of macrolide-decreased susceptibility caused by this efflux pump.

Treatment

Antimicrobial agents regarded as effective for treatment of Legionnaires' disease are macrolides, fluoroquinolones and rifampicin, all achieving adequate intracellular concentrations [10-12]. However, immunocompromised patients may present with slowly or non-resolving disease despite isolation of organism appearing susceptible in vitro [13]. Agents with poor intracellular penetration (e.g. beta-lactam agents and aminoglycosides) are ineffective for treatment despite good in vitro activity.

Antimicrobial susceptibility testing

Currently, there is no gold standard for antimicrobial susceptibility testing (AST) of *L. pneumophila*. Different methods, particularly with different media, yield different MICs [14]. The use of solid media containing charcoal for AST has been subject to debate for a long time and it has been shown that addition of charcoal to the medium will result in elevated MICs for most relevant agents [6,15]. A novel charcoal-free solid media [16] and microbroth dilution method using liquid growth medium [1,2] provide adequate growth of *Legionella* and generate concordant AST results [16]. Recently, an international *Legionella* community (represented by co-authors from various international reference laboratories, countries and clinical stakeholders for diagnosis and treatment of Legionellosis), set out advice for the standardisation of antimicrobial susceptibility testing methods, as well as guidelines and reference strains to facilitate and improve the detection of antimicrobial resistance [14].

As correlation between MICs obtained by any method and clinical outcome is uncertain, AST should be used for the purpose of detecting resistance mechanisms as indicated by MICs above the epidemiological cutoff values (ECOFFs). Sufficient data to establish ECOFFs are not currently available but the published MIC distributions reproduced in the Table 1 and 2 give an indication of MICs for wild type isolates by the two methods presented in this document. The first MIC method uses gradient tests on buffered charcoal yeast extract agar supplemented with α-ketoglutarate (BCYE-α) which provides an easy, reproducible and readily available method for routine laboratories [5,6,15]. The second method, broth microdilution [1,2] is more time-consuming than gradient tests but provides unbiased MICs results due to the absence of charcoal.

1. Gradient test

1. Subculture the strain on BCYE-α medium and incubate for 48 hours at 35-37°C in a humidified atmosphere (50-70% relative humidity).
2. Suspend colonies in sterile water to the density of a 0.5 McFarland standard.
3. Inoculate BCYE-α plates by evenly swabbing the entire surface and apply a single gradient strip to each 9 cm plate.
4. Incubate for 48 hours at 35°C in a humidified atmosphere. If there is inadequate growth after 48 hours, the plates can be incubated for an additional 24 hours.
5. Read MICs at the point of intersection of the ellipse with the gradient strip.
6. Compare the MIC for the isolate with the MIC distributions in the table. The likely susceptibility of the isolate may be reported based on whether the MIC is below or equal to the tentative highest MIC for the wild-type population.
7. Isolates with MICs above the highest MIC for wild-type organisms in the table should be sent to a reference laboratory for confirmation testing

Quality control target MICs and ranges have not been established by EUCAST. However, MICs for *L. pneumophila* NCTC 12821, ATCC 33152, and CIP 107629T should be within the wild type [1,6] and should be used to ensure that growth conditions are adequate. For quality control of gradient strips, follow the instructions of the manufacturer for control of tests on non-fastidious organisms.

Table 1: MIC distributions for *L. pneumophila* (gradient MIC method)

Antimicrobial agent	n	≤0.008	0.016	0.032	0.064	0.125	0.25	0.5	1	2	4	8	16	≥32	Reference[1]
Azithromycin	183			3	43	108	18	7	3			1			[6]
	100		8	31	43	1	8	6			3				[5]
	122			13	45	41	2	16	5						[4]
Clarithromycin	183				1	110	70	1	1						[6]
	100		1	57	20	16	3	1						2	[5]
Erythromycin	183			2	30	119	25	5	1	1					[6]
	100			2	32	57	6	1					2		[5]
	122				12	46	31	30	3						[4]
Ciprofloxacin	183						2	177	3	1					[6]
	100				3	35	57	2	1	2					[5]
	122						24	90	8						[4]
Levofloxacin	183				2	1	177	2	1						[6]
	100			24	63	10		1	2						[5]
	122					43	79								[4]
Moxifloxacin	183						2	176	5						[6]
	100					8	87	2	1		2				[5]
	122						9	70	43						[4]
Rifampicin	183	7	132	39											[6]
	100		97	1						1	1				[5]
	122	57	60	5											[4]
Doxycycline	183								5	48	106	24			[6]
	100			1			1	4	20	69	3	2			[5]
Tigecycline	183								1	15	82	80	5		[6]
	100						1	14	57	21	7				[5]

Note 1: Reference 15 included serogroup 1 only, reference 19 included serogroups 1, 6, 8 and 10, reference 19 included serogroups 1, 3, 4, 5, 6 and 10. MICs may be higher for some sequence types of serogroup 1 than for other serogroups [4,5]

Note 2: Grey shading = tentative highest MIC for wild type organisms

2. Microbroth dilution method

1. Subculture the strain on BCYE-α medium and incubate for 48 hours at 35-37°C in a humidified atmosphere (50-70% relative humidity).
2. Suspend colonies in Liquid growth medium (LGM) to the density of a 0.5 McFarland standard.
3. Inoculate wells containing 40 µL of antibiotic solution with 160 µL of bacterial suspension to give a final bacterial concentration of 4.105 to 5.105/mL
4. Incubate for 48 hours at 35°C in a humidified atmosphere.
5. Read MICs according to the EUCAST recommendations (EUCAST Reading guide, v2.0, 2 March, 2020).
6. Compare the MIC for the isolate with the MIC distributions in the table. The likely susceptibility of the isolate may be reported based on whether the MIC is below or equal to the tentative highest MIC for the wild-type population.
7. Isolates with MICs above the highest MIC for wild type organisms in the table should be sent to a reference laboratory for confirmation testing.

Quality control target MICs and ranges have not been established by EUCAST. However, MICs for *L. pneumophila* NCTC 12821, ATCC 33152, and CIP 107629T should be within the wild type [1,6] and should be used to ensure that growth conditions are adequate. MICs for *Escherichia coli* ATCC 25922 and *Staphylococcus aureus* ATCC 29313 in Liquid growth medium should be similar in Mueller-Hinton broth.

Table 2: MIC distributions for *L. pneumophila* (microbroth dilution method)

Antimicrobial agent	Isolates	n	≤0.002	0.004	0.008	0.016	0.032	0.064	0.125	0.25	0.5	1	2	4	8	16	32	64	128	256	512	1024	2048	Reference
Azithromycin	WT Lp1 LpeAB-neg	93				1	27	53	12															[1]
	WT Lp1 LpeAB-pos	16								2	8	5	1											[1]
	Resistant Lp1*	12															1	1	1	3		2	4	[1]
	WT Lp	40			4	13	8	12	3															[16]
Clarithromycin	WT Lp1	109		1	3	32	61	13																[1]
	Resistant Lp1*	12																1	1	6	4			[1]
Erythromycin	WT Lp1	109					1	20	47	28	9	5												[1]
	Resistant Lp1*	12																				8	4	[1]
	WT Lp	92						14	27	24	26	1												[2]
Ciprofloxacin	WT Lp1	109			9	54	45	2																[1]
	Resistant Lp1*	7								1			2		4									[1]
	WT Lp	92		1	23	58	8			2														[2]
Levofloxacin	WT Lp1	109		1	10	75	24																	[1]
	Resistant Lp1*	7							1			2		4										[1]
	WT Lp	40				9	30	1																[16]
	WT Lp	92					7	62	22	1														[2]
Moxifloxacin	WT Lp1	109			5	29	67	9																[1]
	Resistant Lp1*	7							1		1	1		4										[1]
	WT Lp	92					1	14	75	2														[2]
Rifampicin	WT Lp1	109	109																					[1]
	Resistant Lp1*	10									1	1				3	3	2						[1]
	WT Lp	40	2	28	10																			[16]
	WT Lp	92	92																					[2]
Doxycycline	WT Lp1	109							1	1	7	55	46											[1]
	WT Lp	40											1		6	26	7							[16]

Note 1: Lp: *Legionella pneumophila*; Lp1: *Legionella pneumophila* serogroup 1.

Note 2: * Resistant mutants selected *in vitro* [1,17]

Note 3: Two other refs [18 19] in which the distribution ranges are indicated but results by MIC for each antibiotic are not presented

Note 4: Grey shading = tentative highest MIC for wild type organisms

References

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